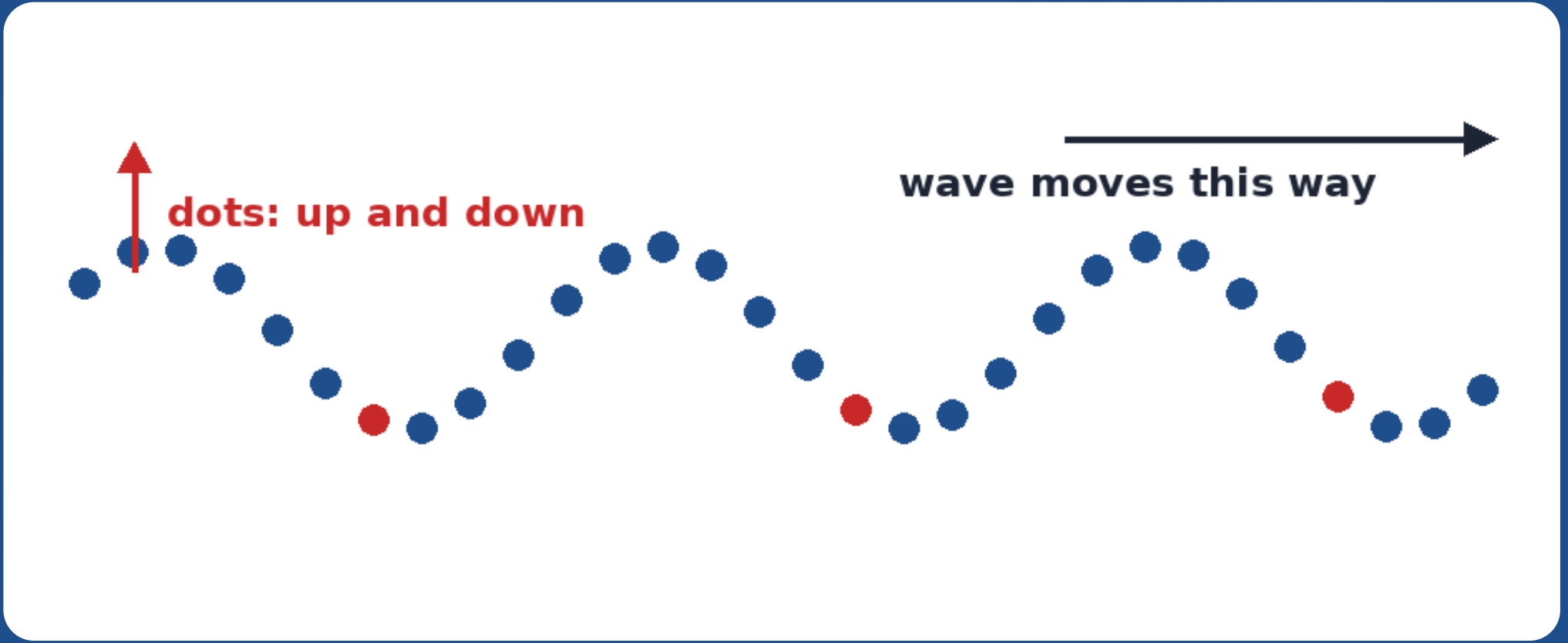
# 4.06 Transverse Waves

How does the medium move in a transverse wave?



## Think About It




**Watch the orange person: up and down only.**



**The crowd does "the wave." What actually travels around the stadium?**

Single choice poll

**A.** the people

Single choice poll

**B.** the wave (energy)

Single choice poll

**C.** the seats

Single choice poll

**D.** nothing moves

# Today's Goals

I can describe how the medium moves in a transverse wave.

I can explain that a wave carries energy, not matter.

I can tell mechanical transverse waves from electromagnetic ones.

I can explain what frequency and amplitude change.

**Rate each goal:**

1 Not yet

2 Getting there

3 I can teach it

# Ohio Science Standards

## 7.PS.4 Energy can be transferred through a variety of ways.

- Mechanical waves need a medium. The medium moves, then returns to its rest position.
- The speed of a wave depends on the material (sound travels faster through most solids than gases).
- Mechanical waves are transverse or longitudinal, depending on how the medium moves.

**Inquiry skills (Grades 5–8):** "Use appropriate mathematics, tools and techniques to gather data" and "Think critically and logically to connect evidence and explanations" (today's JavaLab and toy spring activities).

Source: Ohio's Learning Standards for Science (2018), Grade 7 Physical Science; content statement paraphrased.

## In this lesson

- Integer warm-up
- Vocab check-in
- What is a transverse wave?
- Make a wave (JavaLab)
- Mechanical or electromagnetic?
- Toy spring: frequency and amplitude
- Parts of a wave

# CASE #3 • WEDNESDAY

## Did the coil move 8 cm to the left?

**WAVE WORD: *Longitudinal***

The medium moves back and forth in the same direction the wave travels.

### MYSTERY

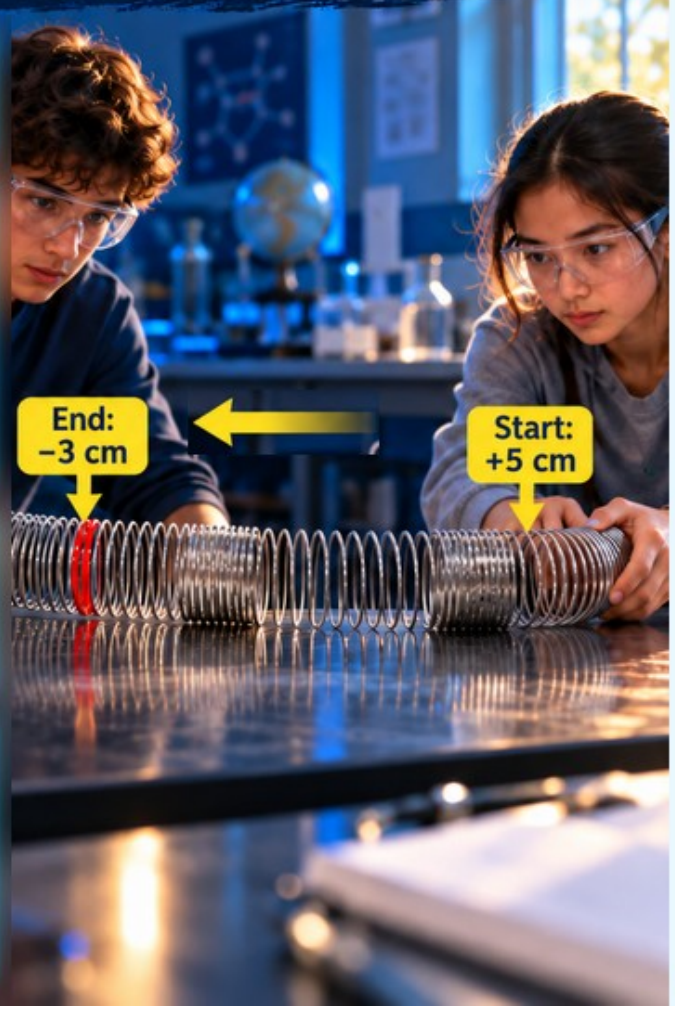
Students push and pull a slinky to make a longitudinal wave. The coils move back and forth in the same direction the wave travels. A painted coil starts 5 cm to the right of its rest spot. Then it moves to 3 cm left of its rest spot. Did it move 8 cm to the left?

### CLUES

- Start: +5 cm
- End: −3 cm
- Change = end − start

**COIL POSITION (cm)**




### CLAIM

**Did the coil move 8 cm to the left?**

☐ YES ☐ NO

### EVIDENCE

**Solve:**

**−3 − 5 = _____**

**Rewrite as adding: −3 + _____ = _____**

### REASONING

**Finish the sentence:**

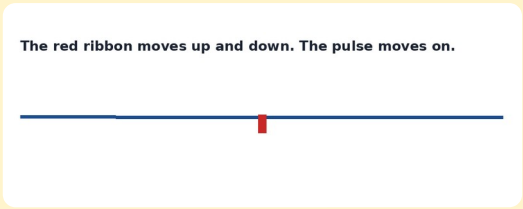
**−3 − 5 is the same as**

**−3 + _____ = _____.**

**So the coil moved _____ cm left.**

# Vocab Check-In

*From your Unit 4 vocabulary pretest and the 4.06 page. Starred words are on the pretest.*




## wave ★

a disturbance that carries energy from one place to another




## medium ★

the material a wave travels through: air, a rope, or a spring



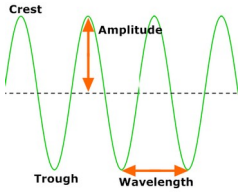

## transverse wave ★

the medium moves up and down, at right angles to the wave



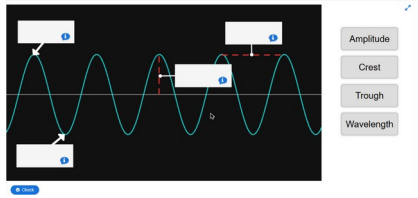

## perpendicular

at a right angle (90°), like the corner of a paper




## crest / trough

the highest point / the lowest point of a transverse wave




## wavelength ★

the distance from one crest to the next crest



## amplitude ★

how far the medium moves from rest; bigger = more energy



## electromagnetic wave ★

needs no medium; light, radio waves, and X-rays

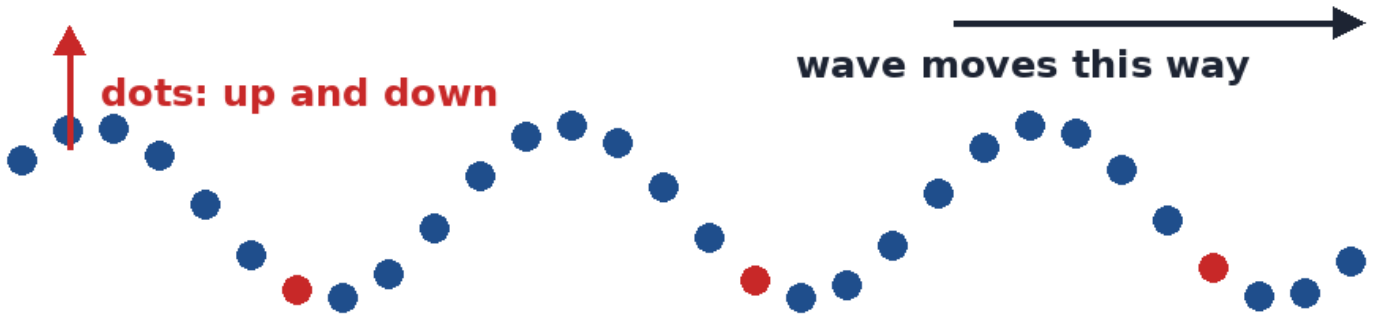
# A Wave Carries Energy, Not Stuff

**The red ribbon moves up and down. The pulse moves on.**



Tie a ribbon to the middle of a rope and flick the end. The ribbon jumps up and drops back down, but it stays in the middle of the rope. The rope (the medium) stays in place. The ENERGY travels down the rope.

# What Is a Transverse Wave?




**Watch the orange person: up and down only.**

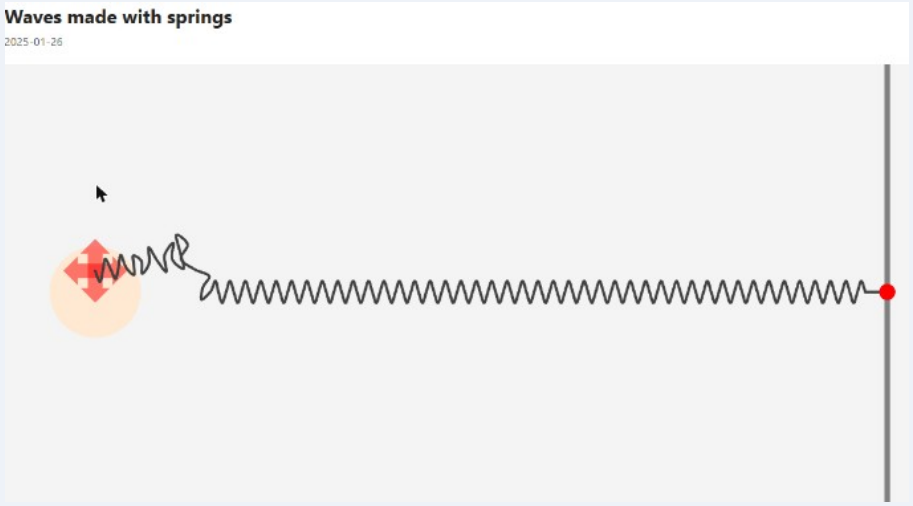


**Transverse wave:** the medium moves up and down, but the wave moves forward.

Those two directions make a **right angle** (they are perpendicular), like the corner of a piece of paper.

Try it: shake a rope tied to a doorknob, or pluck a guitar string.

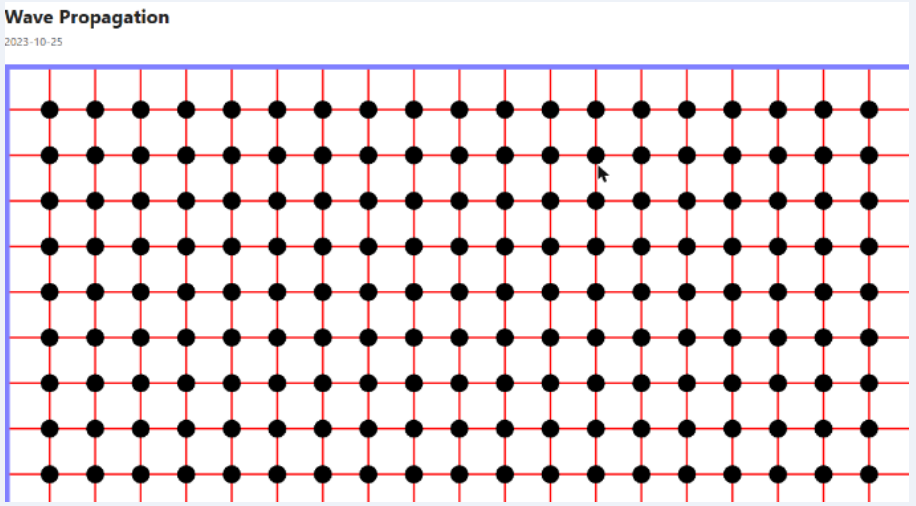
# Part 1: Make a Transverse Wave




## Link 1: Waves made with springs

Click Up-Down. Drag the red handle up and down. Watch ONE coil.

▶️ Open Waves made with springs




## Link 2: Wave Propagation

Check Leaving traces. Drag one dot and let go. Where does each dot end up?

▶️ Open Wave Propagation

# Mechanical or Electromagnetic?

## Mechanical

**needs a medium to travel**

Examples: a rope shaken up and down • a plucked guitar string • a spring shaken up and down

## Electromagnetic (EM)

**does NOT need a medium; can cross empty space**

Examples: sunlight • radio signals • X-rays

Both kinds can be transverse. That is how sunlight crosses almost-empty space to reach Earth, but a wave on a rope cannot.

Engageli Poll

Classkick: 4.06 Part 1 · Q4

# All of these are transverse waves. Which one can travel through empty space?

Single choice poll

**A.** a rope wave

Single choice poll

**B.** a plucked guitar string

Single choice poll

**C.** sunlight

Single choice poll

**D.** a spring shaken up and down

Classkick: 4.06 Part 2 · Q5–Q6

# Part 2: Transverse Waves in a Toy Spring



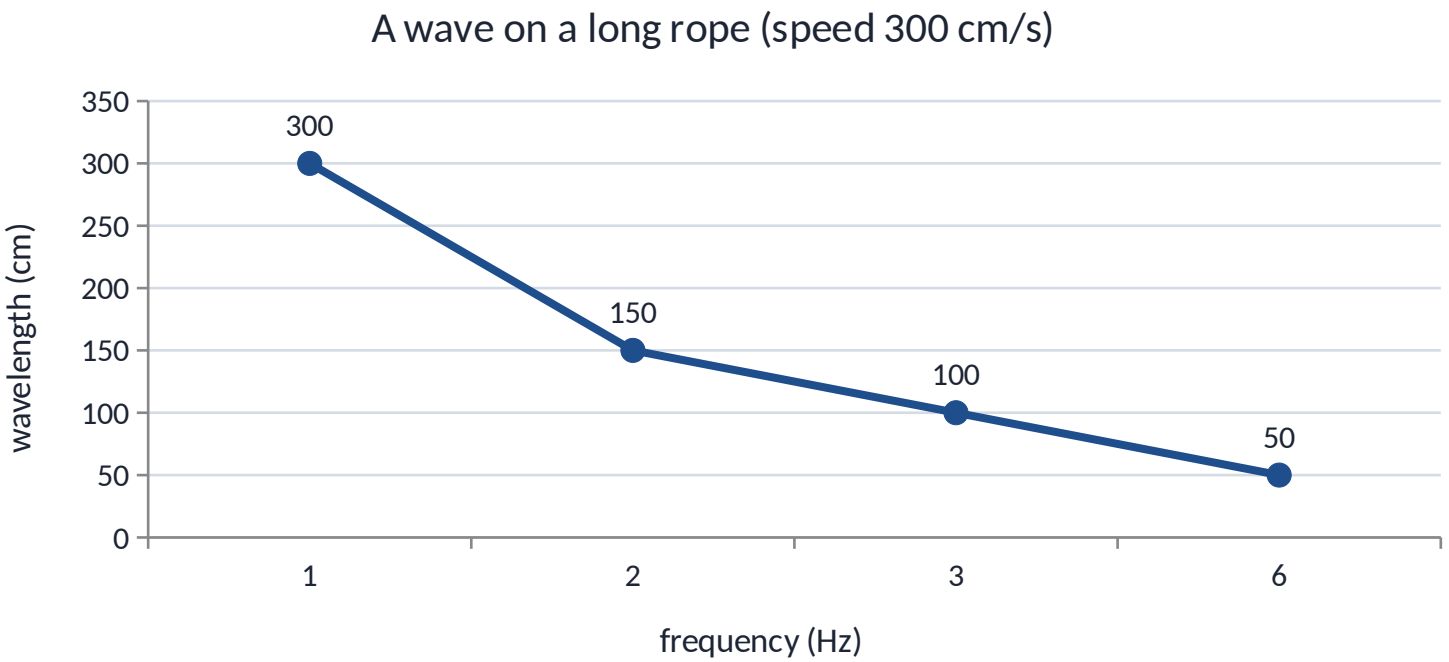

| Frequency | Wavelength |
|---|---|
| 1 Hz | ___ cm |
| 2 Hz | 50 cm |
| 4 Hz | ___ cm |
| 5 Hz | ___ cm |

Move ONLY the Frequency slider. Read the Wavelength under the Pause button. (2 Hz is done for you.)

▶️ Open the toy spring sim

*The shape: an up-and-down S shape with crests and troughs.*

## Graph Poll




Read the graph.

**What happens to the wavelength as the frequency goes up?**

Single choice poll

**A.** It gets longer

Single choice poll

**B.** It gets shorter

Single choice poll

**C.** It stays the same

Single choice poll

**D.** It goes up, then down

# Now Move Only the Amplitude

### Frequency changes…

how close the waves are (the wavelength)

*Higher frequency = shorter wavelength.*

### Amplitude changes…

the height of the wave

*Bigger amplitude = more energy.*

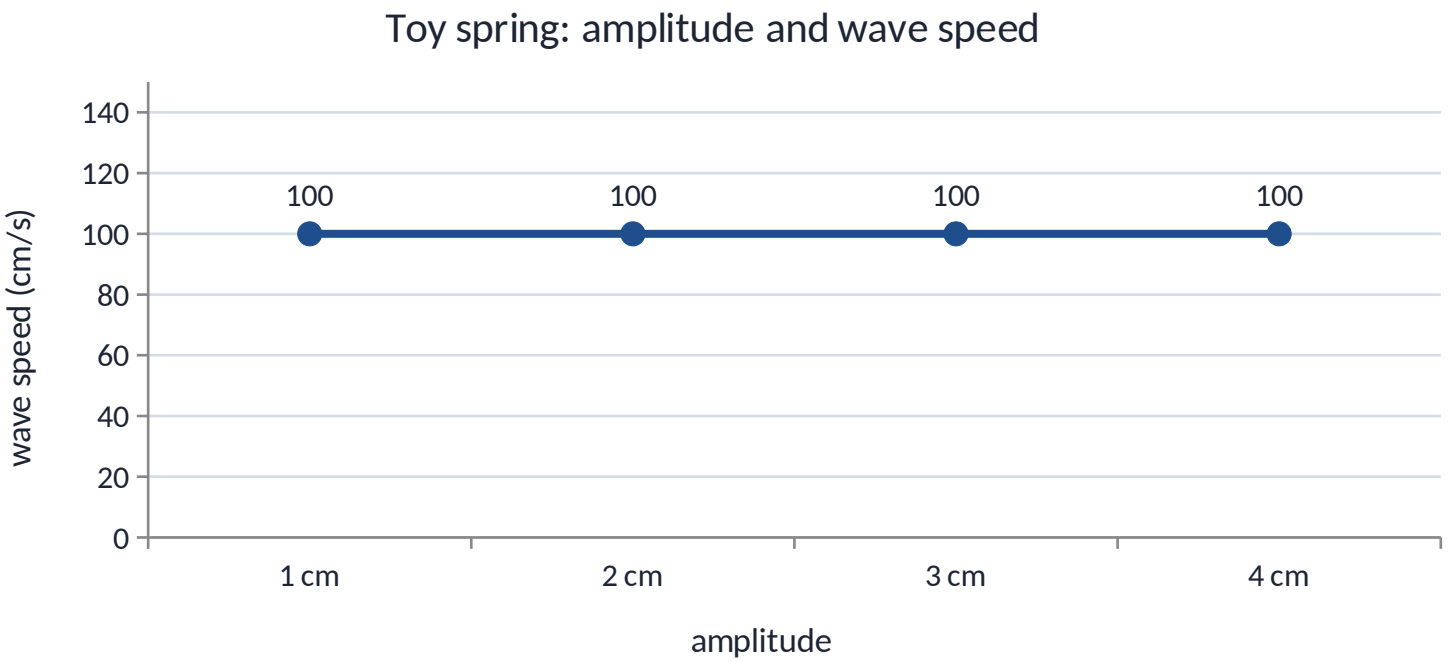
### Neither one changes…

the wave speed (100 cm/s in the toy spring)

*Same medium = same speed.*

▶️ **Open the toy spring sim**

## Graph Poll




Read the graph.

**What does a bigger amplitude do to the wave speed?**

Single choice poll

**A.** Makes it faster

Single choice poll

**B.** Makes it slower

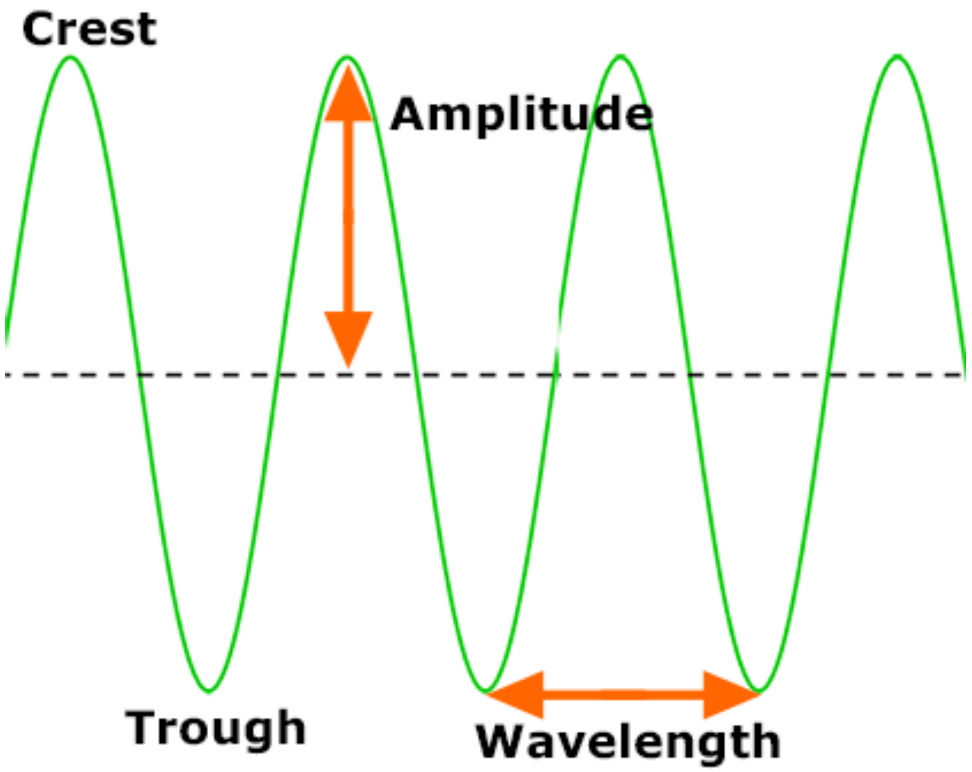
Single choice poll

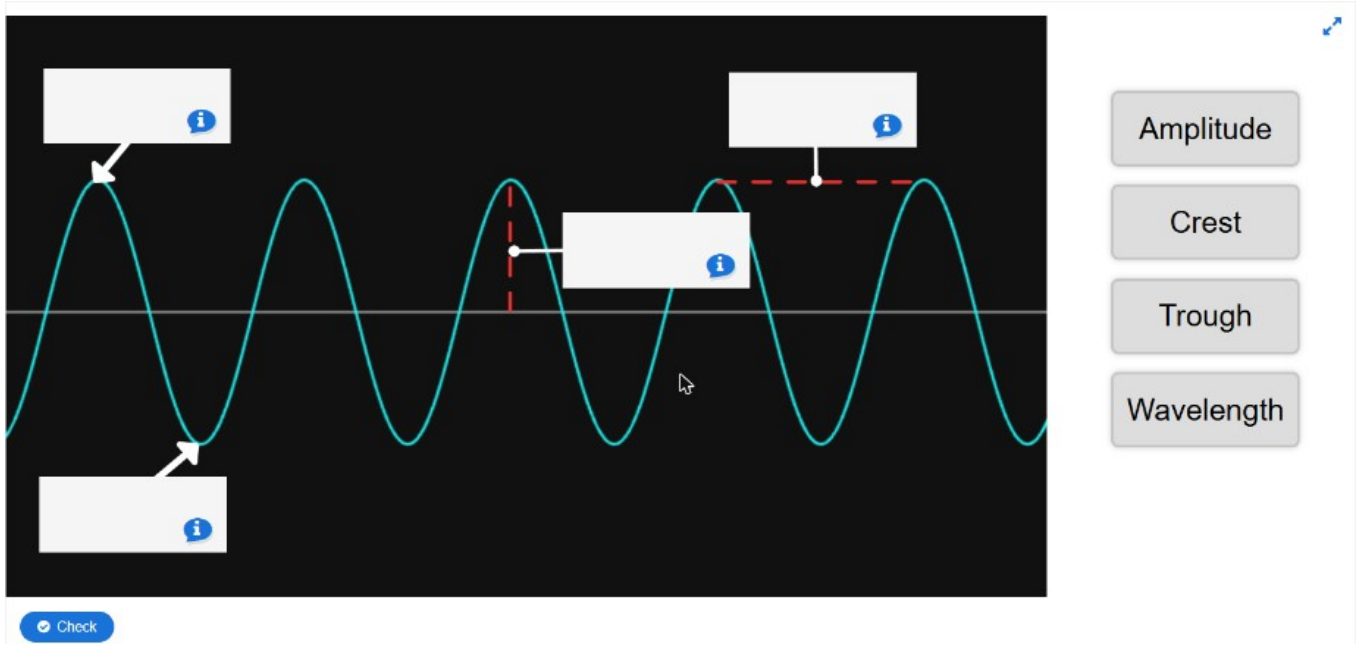
**C.** Nothing: it stays the same

Single choice poll

**D.** Doubles it

# Parts of a Transverse Wave




**Crest** = top. **Trough** = bottom. **Wavelength** = crest to crest. **Amplitude** = rest line to crest.

▶️ **Open the Properties of Waves diagram**

**Who Is Right?**

# A ribbon is tied to the middle of a rope. You flick the rope and a wave passes. Where does the ribbon end up?

### Ava says:

*"At the far end of the rope, because the wave carries it there."*

### Ben says:

*"Back where it started. The rope moves up and down; only the energy travels."*

### Cam says:

*"Higher than before, because the wave lifted it up."*

### Dee says:

*"On the floor, because the wave knocks it off the rope."*



Single choice poll

**A.** Ava



Single choice poll

**B.** Ben



Single choice poll

**C.** Cam



Single choice poll

**D.** Dee

# Key Ideas

1. A wave carries energy, not matter. The medium moves in place.
2. Transverse wave: the medium moves up and down, at right angles to the wave.
3. Transverse waves can be mechanical (rope, guitar string) or electromagnetic (light).
4. Higher frequency = shorter wavelength. Bigger amplitude = more energy. Speed stays the same.

**Re-rate your goals: 1 = Not yet,  2 = Getting there,  3 = I can teach it**